

- 8 The role of sensory receptor cells in mammals is to detect stimuli and generate action potentials in sensory neurones.

Human taste buds on the tongue contain chemoreceptor cells. Different chemoreceptor cells respond to different chemical stimuli.

Fig. 8.1 is a diagram of chemoreceptor cells in a taste bud.

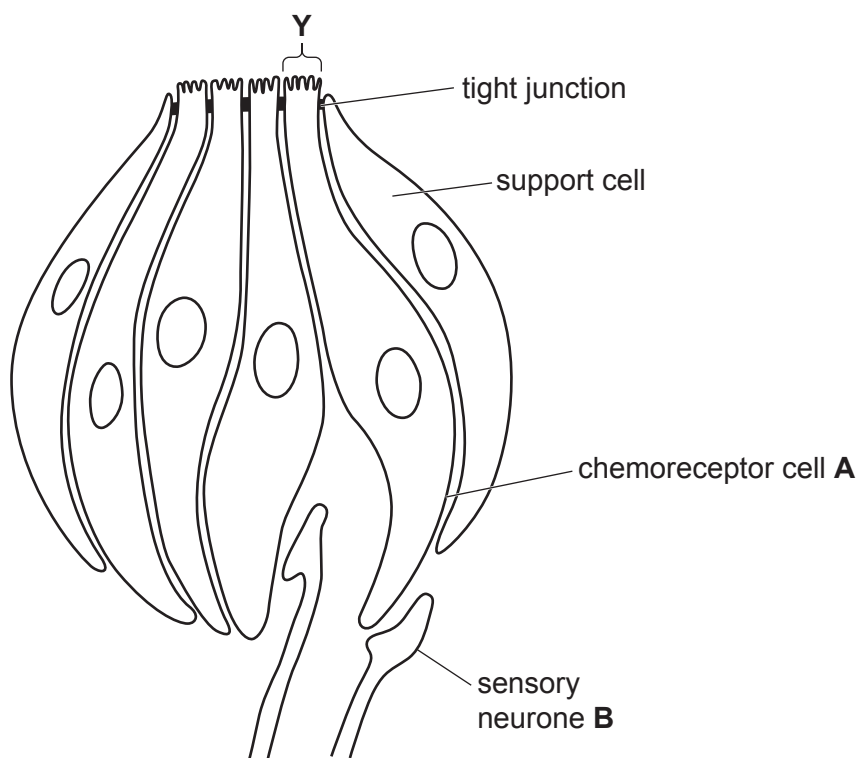


Fig. 8.1

- (a) Name the structures in the region Y.

..... [1]

- (b) Suggest a reason for the tight junctions between the chemoreceptor cells.

.....
.....
..... [1]

- (c)** Chemoreceptor cell **A** responds to sodium ions (Na^+) in salt.

Describe how the contact of cell **A** with Na^+ can result in an action potential in sensory neurone **B**.

[7]

[Total: 9]